

DAY 01

Campusmonk

Campusmonk

Q1. Which of the following is the largest fraction?

$\frac{6}{7}$, $\frac{5}{6}$, $\frac{7}{8}$, $\frac{4}{5}$

a. $\frac{6}{9}$ b. $\frac{4}{5}$ c. $\frac{5}{6}$ d $\frac{7}{8}$

Campusmonk

Q2. Which of the following fraction is the smallest?

$\frac{7}{6}$, $\frac{7}{9}$, $\frac{4}{5}$, $\frac{5}{7}$

a. $\frac{7}{6}$ b. $\frac{7}{9}$ c. $\frac{4}{5}$ d. $\frac{5}{7}$

Campusmonk

Q3. Which of the following is correct?

(1) $\frac{2}{3} < \frac{3}{5} < \frac{11}{15}$

(2) $\frac{3}{5} < \frac{2}{3} < \frac{11}{15}$

(3) $\frac{11}{15} < \frac{3}{5} < \frac{2}{3}$

(4) $\frac{3}{5} < \frac{11}{15} < \frac{2}{3}$

Campusmonk

Q4. Arrange $\frac{4}{5}$, $\frac{7}{8}$, $\frac{6}{7}$, $\frac{5}{6}$ in the ascending order:

a. $\frac{4}{5}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}$

b. $\frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{4}{5}$

c. $\frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$

d. $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}, \frac{6}{7}$

Campusmonk

Q5. The Value of

$$\frac{\frac{2}{7} \times \frac{28}{11} of \frac{121}{14} - \frac{3}{7} of \frac{8}{9}}{\frac{3}{5} \times \frac{15}{8} of \frac{80}{9} - \frac{7}{3} \times \frac{15}{49} of \frac{86}{15}} \; is:$$

TCS NQT 2022

Campusmonk

Q6 TCS NQT 2021

$$1 \div \frac{3}{7} \text{ of } (6 + 8 \times \overline{3 - 2}) + [\frac{1}{5} \div \frac{7}{25} - \{\frac{3}{7} + \frac{8}{14}\}]$$

a. $\frac{5}{42}$ b. $-\frac{5}{42}$

c. 1 d. *None of these*

Campusmonk

Q7. The Value of

$$\left[\frac{(0.63)^2+(0.8)^2+(0.21)^2}{((0.063)^2+(0.08)^2+(0.021)^2)}\right] \div 3\frac{1}{3} \times 5\frac{1}{3} \text{ of } \frac{1}{8}, \text{ is:}$$

TCS NQT 2022

Campusmonk

Q8.

$$13\frac{3}{4} + 17\frac{2}{7} + 31\frac{1}{4} + 15\frac{5}{7} + 12\frac{2}{3} = ?$$

a. $90\frac{3}{4}$ b.$88\frac{3}{2}$ c)$90\frac{2}{3}$

d) None of these

Campusmonk

Q9. Simplified to:

$$99\frac{1}{7} + 99\frac{2}{7} + 99\frac{3}{7} + 99\frac{4}{7} + 99\frac{5}{7} + 99\frac{6}{7}$$

a. 102 b.104 c. 597 d. None of these

Campusmonk

Q10. Simplified to:

$$27\frac{3}{11} + 118\frac{2}{11} - 32\frac{5}{22} = 11\frac{6}{11} + x$$

a. $102\frac{7}{22}$ b.$102\frac{15}{22}$

c) $101\frac{15}{22}$ d) None

Campusmonk

Q11. What value should come in the place of question mark in following question

$$\sqrt{? + 27 * 4 + 119} = 14\frac{2}{3} + 6\frac{1}{3}$$

a. 176 b. 214 c. 252 d. 188 e. 212

Campusmonk

Q12. $\frac{1}{8.33\%}$ of $\frac{810}{4} \div 360 \times \frac{40}{9} = ?$

a. 10
b. 20
c. 30
d. 60
e. NOT

Campusmonk

Q13. $\sqrt{81\%}$ of 800 + $\sqrt[3]{51.2\%}$ of 600 = ?

a. 120 b. 600 c. 720 d. 1200 e. NOT

Q14

$$[(16)^{-1/200\%} - (30\%)^{200\%}]^{50\%} = ?$$

a. 0.04
b. 0.2
c. 0.4
d. 0.16
e NOT

Campusmonk

Q15. The value of

$$1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{2}{3}}}}}$$

a. $\frac{21}{13}$ b. $\frac{17}{2}$ c. $\frac{34}{21}$ d. $\frac{8}{5}$

Q16 find the value of $2x + \frac{7}{4}$

Campusmonk

If $x = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{2}}}}$

$2 \times \frac{13}{8} + \frac{7}{4} = \frac{20}{4} = 5$

a. 5

b. $\frac{13}{8}$

c. 4

d. $\frac{7}{8}$

Campusmonk

Q17 . Find the value of

$$1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{1}{1-\frac{2}{3}}}} =$$

-2 -1

2 3 1 -2 -3 -1

-3 -(-2)

$\neq 1$

$\neq 3$

a. $\frac{2}{3}$ b. $-\frac{1}{3}$ c. $\frac{1}{3}$ d. $-\frac{2}{3}$

Campusmonk

Q18. The value of

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\ldots\ldots\ldots\ldots\left(1+\frac{1}{120}\right)$$

a. 30 b. 40.5 c. 60.5 d.121

$$\frac{3}{2}\times\frac{4}{3}\times\frac{5}{4}\cdots\cdots\cdots\frac{119}{118}\times\frac{120}{119}\times\frac{121}{120}$$

$$\frac{121}{2}=60.5$$

Campusmonk

Q19. The simplified value of

$$\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\ldots\left(1-\frac{1}{99}\right)\left(1-\frac{1}{100}\right)$$

$a.\ \frac{2}{99}$ b. $\frac{1}{25}$ c. $\frac{1}{50}$ d. $\frac{1}{100}$

Hint.

Q20. $\left(1+\frac{1}{2}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{6}\right)\left(1+\frac{1}{8}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)\left(1-\frac{1}{7}\right)$ $= \left(1+\frac{1}{x}\right)$ then what is the value of x?

a. 6 b. 8 c. 5 d. 7

Campusmonk

Q21. Consider a sequence, where the nth term is

$t_n = \frac{n}{(n+2)}$, n=1, 2..

The value of $t_3 * t_4 * t_5 * \ldots t_{53}$ equals

CAT/TCS digital

a. $\frac{2}{495}$ b. $\frac{2}{477}$ c. $\frac{12}{55}$

d. $\frac{1}{1485}$ e. $\frac{1}{2970}$

H.W.

Campusmonk

Q22. The sum of the first 20 terms of the series

$$\frac{1}{5\times6}+\frac{1}{6\times7}+\frac{1}{7\times8}+\ldots\ldots+\frac{1}{24\times25}$$ is:

$\frac{1}{5}-\frac{1}{6}+\frac{1}{6}-\frac{1}{7}+\frac{1}{7}-\frac{1}{8}+\cdots\cdots-\frac{1}{24}+\frac{1}{25}$

$\frac{1}{5}-\frac{1}{25}=\frac{4\times4}{25\times4}$

$\frac{1}{5\times4}=\frac{1}{5}-\frac{1}{6}$

a. 1.6
b. 0.16 ✓
c. 0.016
d. 0.0016

b $\frac{16}{100}$

$1 = 1$

$1 + 1 = 1 \times 2$

$1 + 1 + 1 = 1 \times 3$

$1 + 1 + \ldots = 1 \times n$

Q23. Find sum of

$$\left(1 - \frac{1}{n+1}\right) + \left(1 - \frac{2}{n+1}\right) + \left(1 - \frac{3}{n+1}\right) + \ldots\ldots + \left(1 - \frac{n}{n+1}\right)$$

a. n b. $\frac{1}{2}$n c. (n+1) d. $\frac{1}{2}(n+1)$

$$[1 + 1 + 1 + \ldots n] - \frac{1}{n+1}\,[1 + 2 + 3 + \ldots n]$$

$$n - \frac{1}{n+1}\left[\frac{1}{2}n(n+1)\right]$$

$$n - \frac{n}{2} = \frac{n}{2}$$

Campusmonk

n = 20 pages

1 2 3 4 5 6 7 8 9 10

Sum = $\frac{1}{2} n(n+1) = \frac{1}{2} \times 20 \times 21$

= 210

210 − 195

= 15

"Question of the day"

Q24. One page is torn from a booklet whose pages are numbered in the usual manner starting from the first page as 1. The sum of the numbers on the remaining pages is 195. The torn page contains which of the following numbers?

T CS Digital/Elitmus

a) 5, 6

b) 7, 8

c) 9, 10

d) 11, 12

$\frac{1}{2} \times 19 \times 20^{10}$

190

xx

elitmus

[CSAT]

CAT

MBA

Plant

Campusmonk

$1+2+3 = \frac{1}{2}n(n+1)$

$= \frac{1}{2} \times 3 \times 4^2 = 6$

Q25.

Find the Average of :

1 , 2 , 2 , 3, 3 , 3 , 4, 4, 4, 4, ...10,10,10,10,10(10 times)

6

1 , 2 , 2 , 3 , 3 , 3 , 4 , 4 , 4 , 4 , 5 , 5 ——

$\frac{1}{6}n(n+1)(2n+1)$

$1+4+9+16+25+\cdots-100$

$Avg = \frac{Sum}{No} =$

$\frac{1}{2}n(n+1)$ ← $1+2+3+4+5+\cdots-10$

Comments:

Campusmonk

Q26.

Which value among 3^{200}, 2^{300} and 7^{100} is the largest?

a. 3^{200} **b. 2^{300}** **c. 7^{100}** **d. All area equal**

Campusmonk

Q27. The greatest number among 2^{60}, 3^{48}, 4^{36}, and 5^{24} is:

a. 2^{60} b. 3^{48} c. 4^{36}, d. 5^{24}

Campusmonk

Q28. Which one among $\sqrt[3]{6}$, $\sqrt[2]{5}$ and $\sqrt[6]{12}$ is the largest?

a. $\sqrt[3]{6}$ **b. $\sqrt[2]{5}$** **c. $\sqrt[6]{12}$** **d. All are equal**

Campusmonk

Q29. Which value among $\sqrt[4]{7}$, $\sqrt[3]{11}$ and $\sqrt[12]{1257}$ is the largest?

a. $\sqrt[3]{11}$ b. $\sqrt[4]{7}$ c. $\sqrt[12]{1257}$ d. All are equals

Campusmonk

Q30. The greatest among the numbers $\sqrt[2]{8}, \sqrt[4]{13}, \sqrt[5]{16}, \sqrt[10]{41}$ is:

a. $\sqrt[4]{13}$ b. $\sqrt[5]{16}$ c. $\sqrt[10]{41}$ d. $\sqrt[2]{8}$

Campusmonk

Q31. $\sqrt{121} + \sqrt{12321} + \ldots\ldots\ldots + \sqrt{1234567654321} =$

a. 1234567 b. 1234566 c. 1234565 d. 12345677

Q32. $\sqrt{3\sqrt{3\sqrt{3}\ldots.}}$ is equal to.

a. $\sqrt{3}$ b. 3 c. $2\sqrt{3}$ d. $3\sqrt{3}$

Q.33

$$\left[(68921)^{1/3} - (2744)^{1/3}\right]^{1/3} = ?$$

a. 2
b. 3
c. 4
d. 5
e. *None of these*

Q26. Find the value of $\sqrt{3+\sqrt{3+\sqrt{3+\sqrt{3+\ldots\ldots\infty}}}}$

a. 3 b. 4 c. 6 d. 5